

BRANCH DOCUMENT — IRIS-3

Branch: branch_20260512_010117 | Base: main-v1.0

Generated: 2026-05-12 01:01

Decision: NO_GO | Confidence: 33%

CHANGE REQUEST

delet one solar

CASCADE ISSUES IDENTIFIED

[CRITICAL] ISSUE-001 — Power generation reduction

Subsystem: solar | Req: L1-PWR-002 | Type: constraint_violation

Removing one solar panel reduces the total power generation capacity, potentially exceeding the average power consumption during eclipse and peak imaging mode requirements.

[MAJOR] ISSUE-002 — Structural balance disruption

Subsystem: mechanical | Req: None | Type: missing_test

The removal of one solar panel disrupts the structural balance and symmetry, potentially affecting the payload's dynamic response to launch vibrations.

[MINOR] ISSUE-003 — Thermal imbalance

Subsystem: thermal | Req: None | Type: reanalysis_required

The removal of one solar panel may cause thermal imbalances, affecting the temperature stability and control across orbital cycles.

MISSING ECSS TESTS — NOW REQUIRED

ECSS-E-ST-32C Rev.1 §9 — Shock test to validate structural integrity after solar panel removal

Reason: The change affects the mechanical subsystem's balance and requires revalidation of shock response.

ECSS-E-ST-31C §5.4 — Thermal balance test to validate thermal stability after solar panel removal

Reason: The change may introduce thermal imbalances that need verification through a new thermal balance test.

SUBSYSTEM STATUS — UPDATED

Subsystem	Score	Status	Detail	Changed
optical	92%	OK	MTF 0.42 > threshold 0.40	—
thermal	53%	WARN	Thermal imbalance	■ MODIFIED
mechanical	68%	WARN	Structural balance disruption	■ MODIFIED
structure	91%	OK	Mass 33.8 kg / budget 35 kg	—
solar	54%	FAIL	Power generation reduction	■ MODIFIED
startracker	67%	WARN	Stray light analysis pending	—
dataproc	71%	WARN	Compression ratio TBC	—

AFFECTED REQUIREMENTS

- L1-PWR-002 — verification evidence invalidated, re-analysis required
- L1-MEC-003 — verification evidence invalidated, re-analysis required